

Hazel Bamford

Master's student of Polar and Alpine Change Msc(Res), University of sheffield

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Summary

Msc(Res) candidate: Polar and Alpine Change. Recent Environmental science BSc graduate with a strong foundation in numerical modelling, programming using Python and R with high performance computers. Highly motivated to continue to learn and pursue a career in climate science.

Education

Master of science by research: *Polar and Alpine Change*

University of Sheffield

Dissertation: Glacial meltwater contribution to high Andean wetlands

Bachelor of science: Environmental science

University of York: 2:1

Dissertation: The growth of algae on the Greenland ice sheet

Sir Henry Floyd Grammar School

A levels: Psychology, Chemistry, Biology - **ABB**

Skills

Computing

Python: Iris, Dask, Numpy, Xarray, Pandas, CuPy, Matplotlib, Seaborn

Python Modelling: Open Global Glacier Model (**OGGM**), Coupled Ice and Snowpack model (**COSIPY**)

R: Knitr, Tidyverse, GGplot

Linux/HPC: OpenMPI, SLURM, Git, Bash, Zsh

Familiar with Zarr, Netcdf and Geotiff file formats

Remote Sensing, GIS

Use of Sentinel 2 and Sentinel 5 data

Use of Planet Labs data

Orthorectification and georeferencing of historical aerial imagery

ArcGIS, QGIS

APIs: EarthExplorer M2M, Sentinel Hub, Planet Labs

Word Processing:

Microsoft office suite

Google docs

Latex

Data Analysis

Analysis of bias corrected WRF data and comparison with other available datasets including partial AWS datasets and ERA5 reanalysis data.

Cleaning streamflow data and comparing this to associated AWS data to ensure it appears feasible.

Processing of large satellite imagery datasets - cleaning remote sensing data to perform time series analysis.

Processing of datasets in excess of 100GB

Creation of Plots and animations of time-series data to display in an engaging and informative way

Project Management

I have held a part time job alongside my studies, showing my ability to manage my time effectively and prioritise tasks

Cooperation

My MSc project is part of a wider research project, and discussing my findings and issues has been a key part of my learning

Discussions with international researchers to access data used in their publications, or to discuss their findings.

Collaboration with the developers of glacier models

Being part of departmental seminars has given key understanding of research as a career, and the importance of sharing ideas and working collaboratively

Departmental coffee meetings have allowed me to talk to a wide variety of people and opened opportunities that would not have been available otherwise

Funding

Sheffield Postgraduate Scholarship - £10,000

Fieldtrip Scholarship: £2000

Planet Labs private sector satellite imagery - ~3000 Km² per month for 36 months

Referees

Dr Jeremy Ely, *Senior Lecturer in Physical Geography*

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Dr Robert Bryant *Senior Lecturer and Reader in Earth Observation*

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